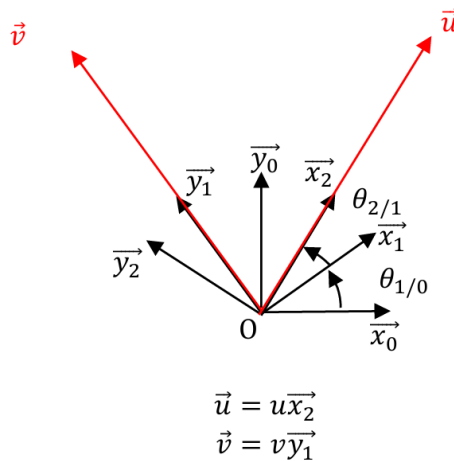


Exercice 1 :



Question 1: Expliciter l'angle orienté $(\vec{x}_2, \vec{y}_1)$ en fonction des angles proposés

$$(\vec{x}_2, \vec{y}_1) = (\vec{x}_2, \vec{x}_1) + (\vec{x}_1, \vec{y}_1) = \theta_{1/2} + \frac{\pi}{2}$$

Question 2: En utilisant la formule de définition, calculer $\vec{u} \cdot \vec{v}$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= u\vec{x}_2 \cdot v\vec{y}_1 = uv\vec{x}_2 \cdot \vec{y}_1 = uv \cos(\vec{x}_2, \vec{y}_1) \\ \vec{u} \cdot \vec{v} &= |uv| \cos\left(\theta_{1/2} + \frac{\pi}{2}\right) = -|uv| \sin \theta_{1/2} = |uv| \sin \theta_{2/1}\end{aligned}$$

Comme précisé en cours, ne pas écrire :

$$\vec{u} \cdot \vec{v} = u\vec{x}_2 \cdot v\vec{y}_1 = |uv| \cos(u\vec{x}_2, v\vec{y}_1)$$

Question 3: En utilisant la formule de définition, calculer $\vec{u} \wedge \vec{v}$

$$\begin{aligned}\vec{u} \wedge \vec{v} &= u\vec{x}_2 \wedge v\vec{y}_1 = uv \sin(\vec{x}_2, \vec{y}_1) \vec{z}_0 \\ \vec{u} \wedge \vec{v} &= uv \sin\left(\theta_{1/2} + \frac{\pi}{2}\right) \vec{z}_0 = uv \cos \theta_{1/2} \vec{z}_0 = uv \cos \theta_{2/1} \vec{z}_0 \\ \vec{z}_0 &= \vec{z}_1 = \vec{z}_2\end{aligned}$$

Comme précisé en cours, ne pas écrire :

$$\vec{u} \cdot \vec{v} = u\vec{x}_2 \wedge v\vec{y}_1 = |uv| \sin(u\vec{x}_2, v\vec{y}_1)$$

Question 4: Projeter $\vec{v}$ dans la base 2

$$\begin{aligned}\vec{v} &= v\vec{y}_1 \\ \vec{y}_1 &= -\sin \theta_{1/2} \vec{x}_2 + \cos \theta_{1/2} \vec{y}_2 \\ \vec{v} &= -v \sin \theta_{1/2} \vec{x}_2 + v \cos \theta_{1/2} \vec{y}_2\end{aligned}$$

Question 5: En faisant intervenir des vecteurs de la même base, calculer $\vec{u} \cdot \vec{v}$

$$\begin{aligned}\vec{u} \cdot \vec{v} &= u\vec{x}_2 \cdot v\vec{y}_1 = u\vec{x}_2 \cdot (-v \sin \theta_{1/2} \vec{x}_2 + v \cos \theta_{1/2} \vec{y}_2) \\ \vec{u} \cdot \vec{v} &= uv(-\sin \theta_{1/2} \vec{x}_2 \cdot \vec{x}_2 + \cos \theta_{1/2} \vec{x}_2 \cdot \vec{y}_2) \\ \vec{u} \cdot \vec{v} &= -uv \sin \theta_{1/2} = uv \sin \theta_{2/1}\end{aligned}$$

Question 6: En faisant intervenir des vecteurs de la même base, calculer $\vec{u} \wedge \vec{v}$

$$\begin{aligned}\vec{u} \wedge \vec{v} &= u\vec{x}_2 \wedge v\vec{y}_1 = u\vec{x}_2 \wedge (-v \sin \theta_{1/2} \vec{x}_2 + v \cos \theta_{1/2} \vec{y}_2) \\ \vec{u} \wedge \vec{v} &= uv(-\sin \theta_{1/2} \vec{x}_2 \wedge \vec{x}_2 + \cos \theta_{1/2} \vec{x}_2 \wedge \vec{y}_2) \\ \vec{u} \wedge \vec{v} &= uv(\cos \theta_{1/2} \vec{x}_2 \wedge \vec{y}_2) = uv \cos \theta_{1/2} \vec{z}_2 = uv \cos \theta_{2/1} \vec{z}_0\end{aligned}$$

Question 7: En utilisant la notation verticale, calculer $\vec{u} \cdot \vec{v}$

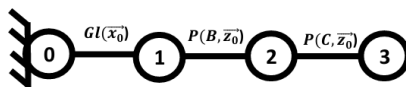
$$\begin{aligned}\vec{u} \cdot \vec{v} &= u\vec{x}_2 \cdot v\vec{y}_1 = u \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}^{B_2} \cdot v \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}^{B_1} = u \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}^{B_2} \cdot v \begin{pmatrix} -\sin \theta_{1/2} \\ \cos \theta_{1/2} \\ 0 \end{pmatrix}^{B_2} \\ \vec{u} \cdot \vec{v} &= -uv \sin \theta_{1/2} = uv \sin \theta_{2/1}\end{aligned}$$

Question 8: En utilisant la notation verticale, calculer $\vec{u} \wedge \vec{v}$

$$\begin{aligned}\vec{u} \wedge \vec{v} &= u\vec{x}_2 \wedge v\vec{y}_1 = u \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}^{B_2} \wedge v \begin{pmatrix} -\sin \theta_{1/2} \\ \cos \theta_{1/2} \\ 0 \end{pmatrix}^{B_2} \\ \vec{u} \wedge \vec{v} &= uv \begin{pmatrix} 0 \\ 0 \\ \cos \theta_{1/2} \end{pmatrix}^{B_2} = uv \cos \theta_{1/2} \vec{z}_2 = uv \cos \theta_{2/1} \vec{z}_0\end{aligned}$$

Exercice 2 :

Question 1: Etablir le graphe des liaisons du mécanisme.



Question 2: Donner l'expression des vecteurs rotation associés aux différentes liaisons pivots

$$\begin{aligned}\vec{\Omega}_{2/1} &= \dot{\theta}_{2/1} \vec{z}_0 \\ \vec{\Omega}_{3/2} &= \dot{\theta}_{3/2} \vec{z}_0\end{aligned}$$

Question 3: Calculer la vitesse $\vec{V}(D, 3/0)$ à l'aide de la définition.

$$\begin{aligned}\vec{V}(D, 3/0) &= \frac{d\vec{OD}}{dt} \Big|_0 = \frac{d(\vec{OA} + \vec{AB} + \vec{BC} + \vec{CD})}{dt} \Big|_0 \\ \vec{V}(D, 3/0) &= \frac{d\vec{OA}}{dt} \Big|_0 + \frac{d\vec{AB}}{dt} \Big|_0 + \frac{d\vec{BC}}{dt} \Big|_0 + \frac{d\vec{CD}}{dt} \Big|_0 \\ \vec{V}(D, 3/0) &= \frac{d\lambda_{1/0}\vec{x}_0}{dt} \Big|_0 + \frac{d\alpha\vec{y}_0}{dt} \Big|_0 + \frac{dL_2\vec{x}_2}{dt} \Big|_0 + \frac{dL_3\vec{x}_3}{dt} \Big|_0 \\ \vec{V}(D, 3/0) &= \dot{\lambda}_{1/0}\vec{x}_0 + L_2\dot{\theta}_{2/1}\vec{y}_2 + L_3(\dot{\theta}_{3/2} + \dot{\theta}_{2/1})\vec{y}_3\end{aligned}$$

Question 4: Calculer la vitesse $\vec{V}(D, 3/0)$ par composition du mouvement.

$$\begin{aligned}\vec{V}(D, 3/0) &= \vec{V}(D, 3/2) + \vec{V}(D, 2/1) + \vec{V}(D, 1/0) \\ \vec{V}(D, 3/0) &= \vec{\Omega}_{3/2} \wedge \vec{CD} + \vec{\Omega}_{2/1} \wedge \vec{BD} + \dot{\lambda}_{1/0}\vec{x}_0 \\ \vec{V}(D, 3/0) &= \vec{\Omega}_{3/2} \wedge L_3\vec{x}_3 + \vec{\Omega}_{2/1} \wedge (L_2\vec{x}_2 + L_3\vec{x}_3) + \dot{\lambda}_{1/0}\vec{x}_0 \\ \vec{V}(D, 3/0) &= \dot{\theta}_{3/2}L_3\vec{y}_3 + \dot{\theta}_{2/1}L_2\vec{y}_2 + \dot{\theta}_{2/1}L_3\vec{y}_3 + \dot{\lambda}_{1/0}\vec{x}_0 \\ \vec{V}(D, 3/0) &= (\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3\vec{y}_3 + \dot{\theta}_{2/1}L_2\vec{y}_2 + \dot{\lambda}_{1/0}\vec{x}_0\end{aligned}$$

Question 5: Exprimer la vitesse $\vec{V}(D, 3/0)$ dans la base 0.

$$\begin{aligned}\vec{y}_2 &= -\sin\theta_{2/1}\vec{x}_0 + \cos\theta_{2/1}\vec{y}_0 \\ \vec{y}_3 &= -\sin(\theta_{3/2} + \theta_{2/1})\vec{x}_0 + \cos(\theta_{3/2} + \theta_{2/1})\vec{y}_0 \\ \vec{V}(D, 3/0) &= (\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3\vec{y}_3 + \dot{\theta}_{2/1}L_2\vec{y}_2 + \dot{\lambda}_{1/0}\vec{x}_0 \\ &= (\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3(-\sin(\theta_{3/2} + \theta_{2/1})\vec{x}_0 + \cos(\theta_{3/2} + \theta_{2/1})\vec{y}_0) \\ &\quad + \dot{\theta}_{2/1}L_2(-\sin\theta_{2/1}\vec{x}_0 + \cos\theta_{2/1}\vec{y}_0) + \dot{\lambda}_{1/0}\vec{x}_0 \\ \vec{V}(D, 3/0) &= (-(\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3\sin(\theta_{3/2} + \theta_{2/1}) - \dot{\theta}_{2/1}L_2\sin\theta_{2/1} + \dot{\lambda}_{1/0})\vec{x}_0 \\ &\quad + ((\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3\cos(\theta_{3/2} + \theta_{2/1}) + \dot{\theta}_{2/1}L_2\cos\theta_{2/1})\vec{y}_0\end{aligned}$$

Question 6: Quelle condition doit être respectée si l'on souhaite que le point D ait un déplacement vertical ?

$$-(\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3\sin(\theta_{3/2} + \theta_{2/1}) - \dot{\theta}_{2/1}L_2\sin\theta_{2/1} + \dot{\lambda}_{1/0} = 0$$

Question 7: Calculer l'accélération $\vec{\Gamma}(D, 3/0)$.

$$\begin{aligned}\vec{\Gamma}(D, 3/0) &= \frac{d\vec{V}(D, 3/0)}{dt} \Big|_0 \\ \vec{\Gamma}(D, 3/0) &= \frac{d(\dot{\lambda}_{1/0}\vec{x}_0 + L_2\dot{\theta}_{2/1}\vec{y}_2 + L_3(\dot{\theta}_{3/2} + \dot{\theta}_{2/1})\vec{y}_3)}{dt} \Big|_0 \\ \vec{\Gamma}(D, 3/0) &= \frac{d(\dot{\lambda}_{1/0}\vec{x}_0)}{dt} \Big|_0 + \frac{d(L_2\dot{\theta}_{2/1}\vec{y}_2)}{dt} \Big|_0 + \frac{d(L_3(\dot{\theta}_{3/2} + \dot{\theta}_{2/1})\vec{y}_3)}{dt} \Big|_0 \\ \vec{\Gamma}(D, 3/0) &= \ddot{\lambda}_{1/0}\vec{x}_0 + L_2\ddot{\theta}_{2/1}\vec{y}_2 - L_2\dot{\theta}_{2/1}^2\vec{x}_2 + L_3(\ddot{\theta}_{3/2} + \ddot{\theta}_{2/1})\vec{y}_3 - L_3(\dot{\theta}_{3/2} + \dot{\theta}_{2/1})^2\vec{x}_3\end{aligned}$$

Question 8: Calculer la valeur de $\vec{V}(D, 3/0)$.

$$\begin{aligned}\vec{V}(D, 3/0) &= (\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3\vec{y}_3 + \dot{\theta}_{2/1}L_2\vec{y}_2 + \dot{\lambda}_{1/0}\vec{x}_0 = (\dot{\theta}_{3/2} + \dot{\theta}_{2/1})L_3\vec{y}_3 \\ V &= \dot{\theta}_{3/2}L_3 = \frac{2\pi}{60} * 5000 * 0.5 \approx 500 * 0.5 = 250 \text{ m/s}\end{aligned}$$